

Exercise C1

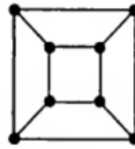
(a) In the Petersen graph, find

1. a trail of length 5;
2. a path of length 9;
3. cycles of lengths 5, 6, 8 and 9;
4. cutsets with three, four and five edges.

(b) Write down $\kappa(G)$ and $\lambda(G)$ for each of the following graphs : C_6 , W_6 , $K_{4,7}$.

(c) Which of the following graphs are Eulerian, semi-Eulerian, Hamiltonian or semi-Hamiltonian?

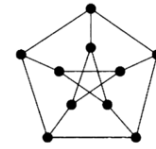
1. The complete graph K_5 ;
2. The complete bipartite graph $K_{2,3}$;
3. The graph of the cube;
4. The graph of the octahedron;
5. The Petersen graph.
6. The Wheel W_6



cube



octahedron



Petersen graph

Exercise C2

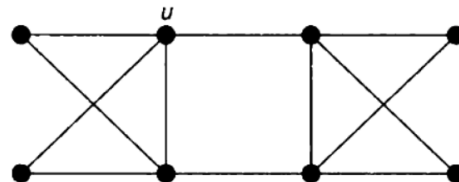
Prove that Any simple graph with n vertices and more than $\frac{(n-1)(n-2)}{2}$ edges, is connected.

Exercise C3

Show, by finding an orientation for each, that K_n ($n \geq 3$) and $K_{r,s}$ ($r, s \geq 2$) are orientable.

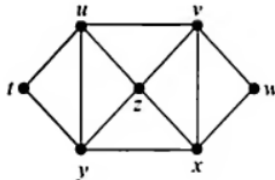
Exercise C4

Use Fleury's algorithm to produce an Eulerian trail for the following graph:

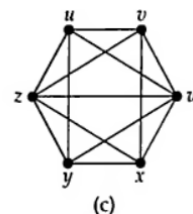
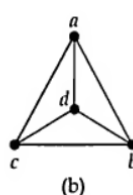
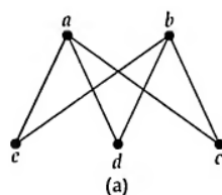


Exercise C5

(a) Show how the following graph can be split into cycles, no two of which have an edge in common.

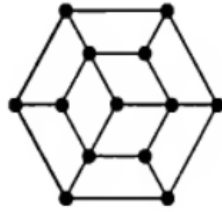


(b) Determine which of the following graphs are semi-Eulerian, and write down a corresponding open trail where possible:



Exercise C4

- (a) Prove that a bipartite graph with an odd number of vertices is not Hamiltonian.
- (b) Use the result of part (a) to prove that the following graph is not Hamiltonian.



Exercise P1

- (a) Show that, if G is a connected graph with minimum degree k then $\gamma(G) < k$.
- (b) Prove that a graph is 2-edge-connected if and only if any two distinct vertices are joined by at least two paths with no edges in common.
- (c) Prove that a graph with at least three vertices is 2-connected if and only if any two distinct vertices are joined by at least two paths with no other vertices in common.

Exercise P2

In a connected graph, the distance $d(v, w)$ between a vertex v and a vertex w is the length of the shortest path from v to w .

- (i) If $d(v, w) \geq 2$, show that there exists a vertex z such that

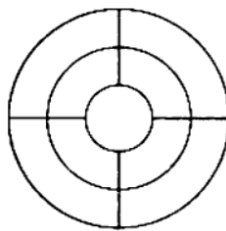
$$d(v, z) + d(z, w) = d(v, w)$$

- (ii) In the Petersen graph, show that $d(v, w) = 1$ or 2 , for any distinct vertices v and w .

Exercise P3

Let G be a connected graph with $k(> 0)$ vertices of odd degree.

- (a) Show that the minimum number of trails, that together include every edge of G and that have no edges in common, is $\frac{k}{2}$.
- (b) How many continuous pen-strokes are needed to draw the diagram in the figure without repeating any line?



Exercise P4

- (a) Give an example to show that the condition $(deg(v) \geq \frac{n}{2})$, in the statement of Dirac's theorem, cannot be replaced by $(deg(v) \geq \frac{n-1}{2})$.
- (b) Let G be a graph with n vertices and $\{\frac{1}{2}(n-1)(n-2)\} + 2$ edges. Use Ore's theorem to prove that G is Hamiltonian.
- (c) **Find a non-Hamiltonian graph with n vertices and $\{\frac{1}{2}(n-1)(n-2)\} + 1$ edges.**